- We see in (Fig b) that starting from number 8, all the terms of the sequence stay within the limits of the "allowed" band, proving the validity of (1) for these terms.
- We, of course, assume that this situation will realize for all $n \geq 40$, i.e. for the whole infinite "tail" of the sequence not shown in the diagram.
- Thus, for the selected ε the number N does exist. In this particular case we found it to be 7.

Reader: Hence, we can regard a as the limit of the sequence.

Author: Don't you hurry. The definition clearly emphasizes: "for any positive ε".

So far we have analyzed only one value of ε. We should take another value of ε and find N not for a larger but for a smaller ε. If for the second ε the search of N is a success, we should take a third, even smaller ε, and then a fourth, still smaller ε, etc. repeating each time the operation of finding N.

- In Fig 7c three situations are drawn up for ε_1, ε_2 and ε_3 (in this case $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$). Correspondingly, three "allowed" bands are plotted on the graph. For a greater clarity, each of these bands has its own starting N. We have chosen $N_1 = 7$, $N_2 = 15$ and $N_3 = 27$.
- Note that for each selected ε we observe the same situation in Fig 7c: up to a certain n, the sequence, speaking figuratively, may be "indisciplined" (in other words, some terms of may fall out of the limits of the corresponding "allowed" band). However, after a certain n is reached, a very rigid law sets in, namely, all the remaining terms of the sequence (their number is infinite) do stay within the band.

Reader: Do we really have to check it for an infinite number of ε values?

Author: Certainly not. Besides it is impossible. We must be sure that whichever value of $\varepsilon > 0$ we take, there is such N after which the whole "infinite tail" of the sequence will get "locked up" within the limits of the corresponding "allowed" band.

Reader: And what if we are not so sure?

Author: If we are not and if one can find a value of ε, such that it is impossible to "lock up" the infinite "tail" of the sequence within the limits of its